Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Applied AI engineer specializing in verified-source document intelligence and forward deployment of production LLM systems for enterprise clients under governance, latency, and reliability constraints. Built **Leasel** — a three-surface enterprise AI platform: Document Intelligence (OCR ingestion, **100% citation correctness**, guardrail system blocking unsupported outputs, human-review gate on regulated decisions), Case Investigation (8-agent LangGraph pipeline with tool use, structured outputs, ranked evidence-backed hypotheses, HITL escalation), and Self-Service Routing (regulated questions always route to staff). Validated with a bespoke evaluation harness: pass@k multi-trial scoring, regression gates, LLM-as-judge quality metrics, and observability across every eval run. 93.6% precision / 87.9% F1 across 1,240 enterprise documents. 8+ years embedding with enterprise compliance, operations, and product stakeholders at Entrata (100K+ MAU) — translating complex regulatory requirements into production systems under strict governance and auditability controls. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Verified-Source Document Intelligence (100% Citation Correctness) | Multi-Agent Architecture, Tool Use & HITL Design (LangGraph, 8-agent) | Forward Deployment & Enterprise Client Delivery | Evaluation Harnesses, Regression Gates & LLM-as-Judge Observability | RAG Pipelines, Guardrails & Grounded Retrieval (ChromaDB, TF-IDF, OCR) | Python · FastAPI · TypeScript · SQL | AWS (S3, EC2, RDS, Glue) · Docker · GitHub Actions CI/CD | Governance, Auditability & Regulated Sector AI (Legal, Financial, Healthcare)

WORK EXPERIENCE

Entrata — PropTech / Real Estate SaaS

Sep 2018 – Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Embedded with compliance, operations, and product teams** to translate complex federal and state regulations into production modules — 20+ full-stack modules across 120+ Agile sprints for 100K+ MAU; zero federal/state compliance violations over 6 years.
- **Shipped Entrata's first production AI pipeline** across 1.6M financial records (Python, PyTorch, OpenAI API) — LLM-powered automation with output validation, quality gates, and compliance controls; zero incidents post-deployment.
- **Built compliance-gated data pipelines** on AWS (S3, Glue, RDS) for regulated financial operations (HAP, HUD, income eligibility, AMI pricing) — optimized 200+ SQL queries for latency and throughput (30% gain); 99.5% uptime over 6+ consecutive months under strict compute and reliability constraints.
- **Achieved 100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines, authentication controls, and auditability logging — zero security incidents. Led Docker microservices redesign: 50% scalability improvement, 30% deployment time reduction.
- Eliminated 80% of manual operational workload via automation; conducted 300+ peer code reviews; mentored 5+ engineers.

George Mason University

Aug 2024 – Aug 2025

Research Associate / Graduate Teaching Assistant

- Built multi-source prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection across 3 research pipelines. Mentored 600+ students in Python and database design.

PROJECTS

Leasel: Verified-Source Legal Document Intelligence & 8-Agent HITL Pipeline

Built for the exact problem law firms face: AI that is grounded in verified sources and never hallucinates. Three surfaces, one knowledge repository: (1) Document Intelligence — OCR ingestion, field extraction, **100% citation correctness** (every answer cited to exact source clause, or explicitly flagged unsupported and downgraded), document diff, missing/conflicting clause detection, human-review gate on all regulated decisions; (2) Case Investigation — 8-agent LangGraph pipeline producing ranked, evidence-backed hypotheses across documents, settings, and historical precedents; conditional HITL escalation when confidence too low; (3) Self-Service Routing — general questions answered directly; anything requiring professional judgement always routes to staff. Core rule enforced in code: LLM extracts and reasons; human makes the final call. **93.6% precision / 87.9% F1** across 10 compliance rules on 1,240 enterprise documents (5 US states + federal).

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, PHP, R, Bash

AI & Agents: LangGraph, LangChain, Multi-Agent Orchestration, Tool Use Pipelines, Structured Outputs, HITL Design, Guardrails, Evaluation Harnesses, LLM Evaluation, LLM-as-Judge, Agent Observability, Prompt Engineering, RAG Pipelines, Vector Search, ChromaDB, Embeddings, OCR, OpenAI API, Anthropic API, FastAPI

ML & Data: PyTorch, Scikit-learn, XGBoost, Isolation Forest, NumPy, Pandas, spaCy, BERT, Hugging Face, Tableau

Cloud & Infra: AWS (S3, EC2, RDS, Glue), Azure, Docker, GitHub Actions CI/CD, MLOps, PostgreSQL, MySQL, ETL, Data Pipelines, Microservices, RabbitMQ